WHEN BACKGROUND NOISE HAS MEANING

Environmental sound classification under semantic interference

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Same target. Same relative level. Different interference.

2,000 ESC-50 clips · official five-fold evaluation · paired mixtures for every model

White noise

Acoustically nonspecific

DEMAND

Traffic · station · metro

Same category

Different fine class

Different category

Different coarse category

+10 dB

Weaker interference

+5 dB

Primary comparison

0 dB

Equal-power stress test

24,000 own-activity mixtures; 23,991 target-activity mixtures after three undefined pairs × three levels are excluded.

Literature and course foundations

Each method addresses a different part of the research question

Course DSP	STFT, MFCCs and spectral descriptors Interpretable measurements and explicit assumptions
Piczak; Salamon & Bello	Learned time-frequency features and augmentation Test whether robustness transfers to competing events
CLAP; PAT	Audio-text representation and readout Freeze the model; separate fixed prompts from full PAT
Research gap	Known distractor identity at controlled power Trace where the decision moves and what remains uncertain

Report Section 2 compares strengths, limitations and the reason for selecting each representation.

Three representations, one controlled experiment

Deployment behaviour is comparable; total pretraining history is not.

A **Handcrafted + SVM**

Training-fold standardisation

STFT → spectral descriptors
MFCCs + deltas → 92 features

B **Compact log-Mel CNN**

Three optimisation seeds

STFT → Mel filters → log power
36,258 trainable parameters

C **Frozen LAION-CLAP**

No training or fine-tuning

Audio / text embedding similarity
50 fixed class descriptions

Custom STFT checked against SciPy and actual course MATLAB code; maximum absolute discrepancy 2.93×10^{-14}.

Power matching depends on where power is measured

The original target-mask definition is retained as a full sensitivity analysis.



Own-activity power

Measure each source over its own active interval.

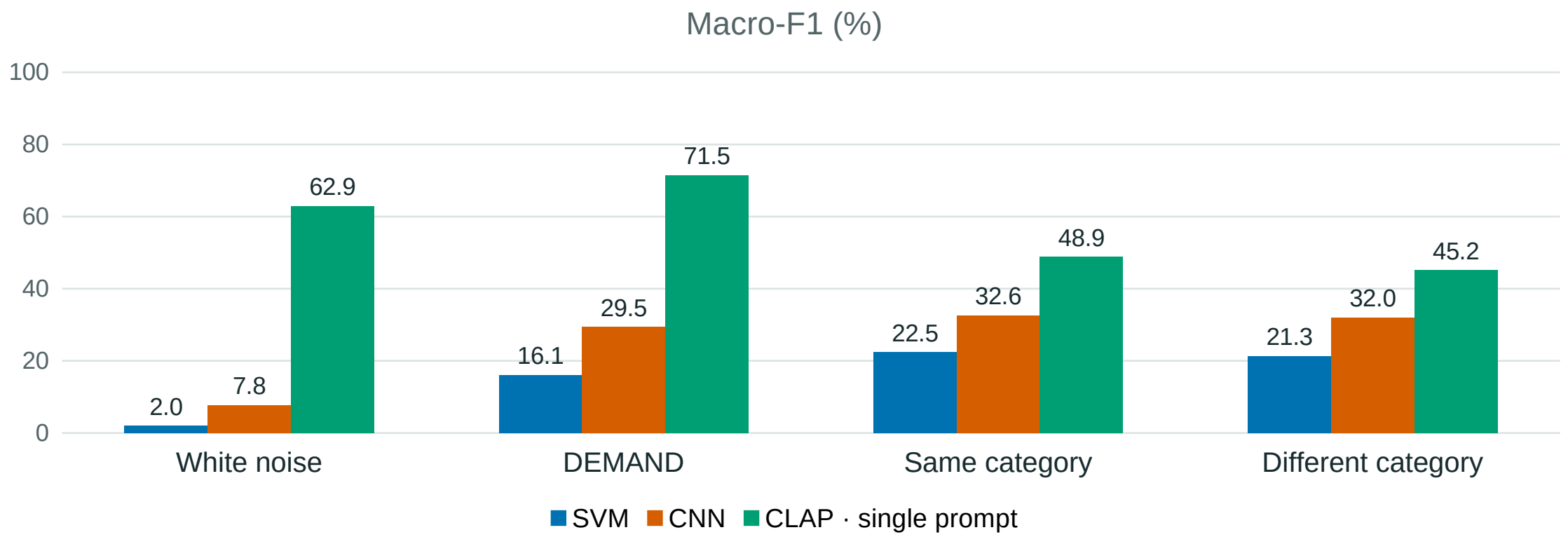
Target-activity power

Measure both sources over the target’s active interval.

Illustrative masks. Common peak attenuation prevents clipping; a gain-matched clean control isolates its effect.

Interference power alone does not predict failure

Own-activity protocol · +5 dB · five-fold mean macro-F1



The report gives fold SD, all three levels, retained-target comparisons and the complete target-activity counterpart.

Does robustness training transfer to competing events?

Same augmentation probability and epoch cap · clean validation · three CNN seeds

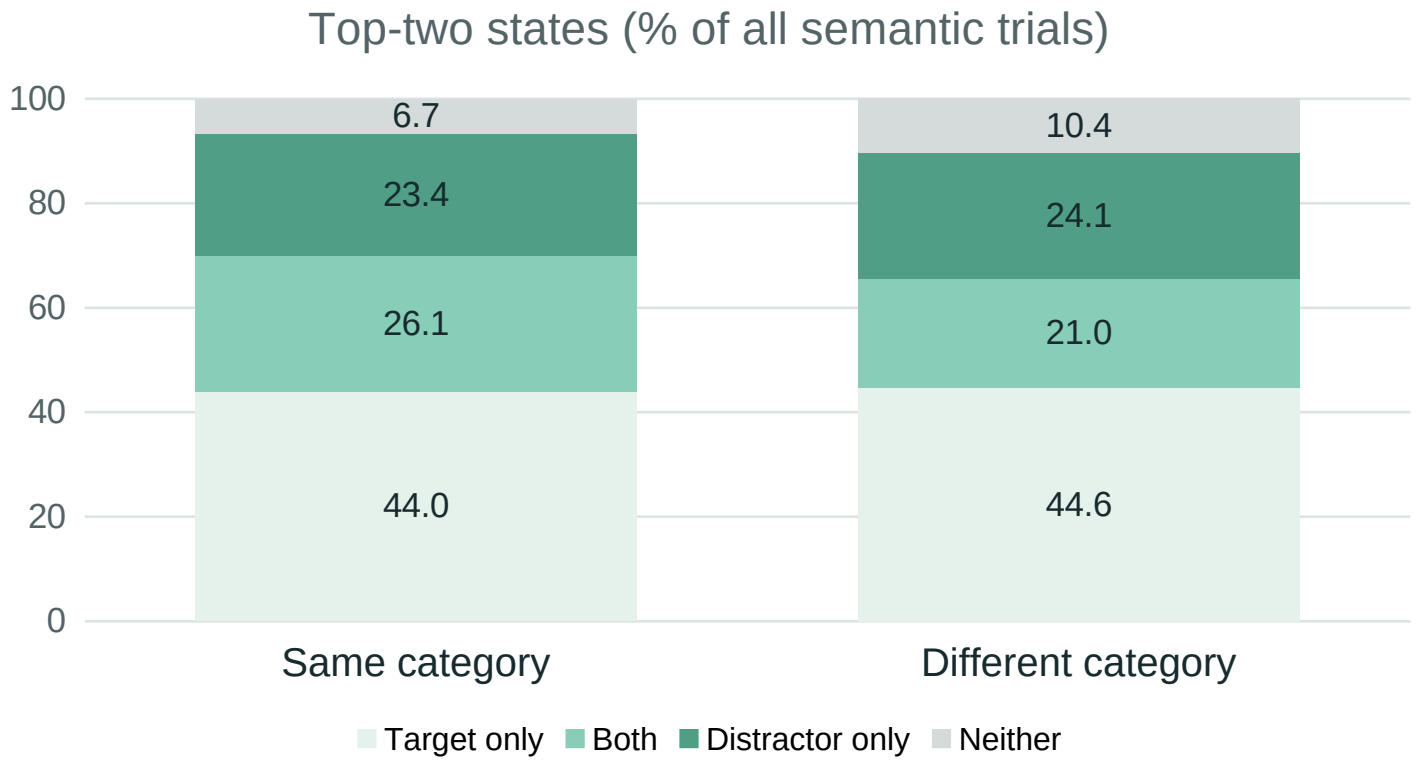
Model / training	Clean	White	DEMAND	Same	Different
SVM / clean	48.0 ± 4.1	2.0 ± 0.9	16.1 ± 2.0	22.5 ± 2.7	21.3 ± 2.6
SVM / WGN	46.0 ± 4.1	41.1 ± 5.4	23.7 ± 5.1	22.6 ± 3.4	22.2 ± 2.6
SVM / semantic	49.3 ± 2.4	1.9 ± 0.6	19.9 ± 2.6	26.8 ± 2.3	25.0 ± 0.9
CNN / clean	65.8 ± 0.7	7.5 ± 0.4	30.9 ± 1.4	33.0 ± 0.3	31.4 ± 0.5
CNN / WGN	64.6 ± 0.6	43.9 ± 2.7	33.8 ± 2.2	33.5 ± 1.0	32.1 ± 1.1
CNN / semantic	61.7 ± 2.4	9.9 ± 0.2	33.6 ± 1.6	36.1 ± 0.8	33.7 ± 0.9

Own activity · macro-F1 (%) · interference conditions at +5 dB

CNN: mean ± seed SD; SVM: mean ± fold SD. Direct CNN semantic-over-WGN accuracy gains do not pass the 156-test Holm correction.

A captured decision can retain the target at rank two

Single-template CLAP · +5 dB · own activity · 2,000 targets per category type



37.34% / 33.17%

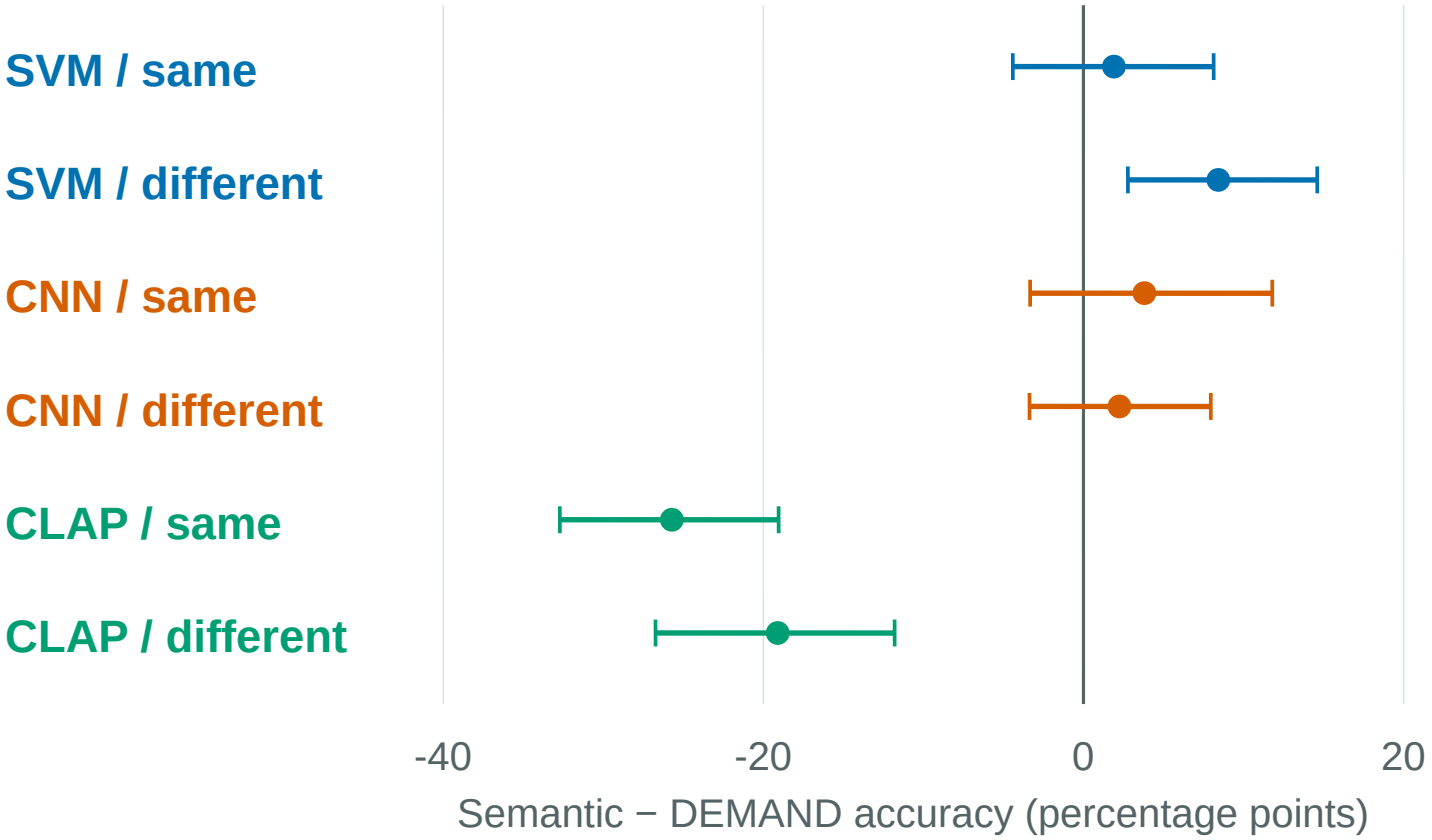
of original captures retain the target at rank two (same / different category)

Top-2 helps some cases. It does not prove that the model heard both sources.

Clean-control joint inclusion: 5.40% / 1.70%. Target-power results and k = 1, 3, 5 sensitivities are retained.

CLAP retains a gap in acoustically close subsets

Same-target comparison · +5 dB · own activity · primary calipers



CNN intervals include zero.
CLAP remains lower.

210 / 178 targets retained
(same / different category)
≈ 9–10.5% of the benchmark

Selected-population association; not proof of semantic causation.

Pointwise 95% target-source bootstrap intervals, 2,000 resamples. Both protocols: CLAP −16 to −26 pp. Subsets differ from high-overlap groups.

Real code execution and sound examples

Can opening + clock alarm · +5 dB SIR · full target-activity overlap

Gain-matched clean control

Play 5-second audio

Target + weaker clock alarm

Play 5-second audio

SVM

can opening
↓
chirping birds

Open the live code demonstration

CNN

can opening
↓
clock alarm

CLAP

can opening
↓
clock alarm

Recorded execution video

Discussion and conclusion

Representation, training and readout explain different parts of the evidence

01 Representation matters

Interference and augmentation affect
the three systems differently.

02 Readout changes the answer

Some argmax captures retain the target
at rank two; many do not.

03 Acoustic controls narrow the claim

CLAP retains a gap in selected subsets;
this does not establish semantic causation.

Limits: three backgrounds · one pair per category type · pretraining overlap · 0-dB label ambiguity

Report, supplement and reproducible code

10-page report · 26-page supplement · working code · execution video

ESC-50 / Piczak	Official folds, dataset caveats and classic spectrogram CNN
LAION-CLAP / Wu et al.	Frozen audio-language reference, ICASSP 2023
PAT / Seth et al.	Released prompt-list ablation; full PAT is not reproduced
Mix-CLAP / Kohno et al.	Noise-aware audio-language model context, ICASSP 2026
RSA-Bench / Zhang et al.	Structured-background speech/reasoning context, ACL Findings 2026

github.com/Charlie-Uni/elec5305-project-550489593

Audio: LG, Can Open.WAV (CC BY); hookhead, alarm clock.aif (CC0). ESC-50 compilation: CC BY-NC 3.0.

AI-assisted implementation and preparation are disclosed. Source audio licences and model provenance remain in the report.